STUDENT IDENTITY CARD GENERATION SYSTEM

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KHYBER PAKHTUNKHWA
PAKISTAN
2022

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A thesis submitted to the department of Computer Science Govt Degree College Wari Dir Upper as a partial fulfillment of the Shaheed Benazir Bhutto University Sheringal Dir Upper for the award of Degree of BS (4-Year) Computer Science.

DEPARTMENT OF COMPUTER SCIENCE
GOVERNMENT DEGREE COLLEGE WARI (DIR UPPER)
KHYBER PAKHTUNKHWA
PAKISTAN
Session 2018-22

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Certificate of Approval

It is certified that the project done in this thesis, entitled **"Student Identity Card Generation System**," was conducted by **Asad Ullah** and **Islam Bahadar** under the supervision of **Dr. Saeed Ullah Jan;** it is further certified that no part of the work has been submitted in another institute for another purpose. This work is submitted to the Government Degree College Wari (Dir Upper), affiliated with Shaheed BB University Sheringal Dir Upper, which partially fulfills the requirements for the degree of BS (4-Year) **Computer Science.**

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Dedication

We dedicate this work to all those who help us conduct this work, especially our parents, teachers, and group members.

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ACKNOWLEDGEMENTS

All praises are for almighty Allah, who bestowed me the potential and capacity to fulfill this challenging job and contribute to the existing knowledge, although in a fraction. All praises are for **Hazrat Muhammad (PBUH)**, a model of guidance and inspiration for humanity.

We want to express my most tremendous gratitude and utmost respect to our advisor Dr. Saeed Ullah Jan, A/P of CS at the Department of Computer Science, Government Degree College Wari, for his enthusiastic guidance, prudent supervision, and continuous encouragement.

Asad Ullah and Islam Bahadar

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PROJECT IN BRIEF

The ID card generator system is an application that will help generate ID cards in an educational institute in government degree college Wari Upper Dir; for ID card generation, there needs to be a proper system. The Administration of government degree collageWari has a manual for the students to fill all the fields in the concerned form. After filling out the form is sent to a computer IT specialist in the market to make the ID card; keeping in view all these issues, I decided to develop an ID card generator system for the government degree college Wari Upper Dir.

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ACRONYMS

CMS	Content Management System
DFD	Data Flow Diagram
ERD	Entity Relationship Diagram
IDE	Integrated Development Environment
HTML	Hyper Text Markup Language
CSS	Cascading Style Sheet
XAMPP	Cross Platform (X), Apache (A), MYSQL (M), PHP
UML	Unified Modeling Language
XML	Extensible Markup Language
API	Application Program Interface
OS	Operating System
DB	Database
DBMS	Database Management System
RDBMS	Relational Database Management System
HTTP	Hyper Text Transfer Protocol

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ABSTRACT

To identify the individual in the educational system, an individual must have an ID card. Not only in educational institutes is also a necessary factor for identification in every field. In an educational institute, the ID card is used to identify the students and employees. Keep in mind that it is necessary to generate the ID cards for students, teachers, and clerical staff of the educational institute. It is today's demand in every educational institute, especially in colleges, because there are more than two thousand students and more than sixty employees in government degree college WariDir Upper. To date traditional system for card generation is in use, which is not adequate; to overcome this, we developed an automated system titled "ID Card generator system" for government degree college Wari Upper Dir.

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CHAPTER – 1

INTRODUCTION

1.1 Introduction

The present-day demand for the great need to measures to ensure the safety as well as security in educational institutes; their educational campuses, including schools, colleges, and universities, need to assure definite well-founded conditions and environment to ensure such type of conditions in educational institutes cards for the students, teachers, and visitors. For this purpose, the correct assignment for the ID card system is essential. This issue can be solved by using up online ID card generator system. Government degree college is a government college located in Wari District Dir Upper in government degree college. Wari, almost two thousand and five hundred students are studying in different departments in different semesters. There is a traditional system for a government degree college wari for the ID card. In the conventional method, at the time of admission, an extra form is given to the students in which they put their detail according to the regarded information in the state after providing the information. The head of the department of each department sends the form to the clerical staff. The clerical staff then sends the received form to the market to make student ID cards. The traditional process of ID card making is time-consuming as well as expensive. To reduce the time consumption and cost of making ID cards for the students of government degree college Wari. I develop a student id generator system for government degree college Wari. The proposed project will solve the issue in the existing process by using the online ID card generator system with the minimal time taken and fast processing.

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1.2Problem Statement

ID card generator system is an application that is used to transmit the details provided by someone into digitized ID cards. In the government degree college Wari, it comes to know that most students providing the information and collecting ID cards waited for several hours. It also shows that most students waste the whole day in this process; hence, the proposed ID Card generator system for government degree college Wari will overcome these issues.

1.3 Aims and objectives

The project has the following aims and objectives, which are enlisted below:

- To develop an ID card generator system for government degree college Wari Upper Dir.
- To develop such a system which speeds up the ID card-making system for the students of government degree collage Wari Upper Dir

1.4Proposed system

Our student ID card generator system will generate Identity cards for students of government degree colleges with many templates to choose from. There will be a database of all college students and an interface that will quickly generate ID cards in a matter of seconds. Project Description

Introduction

On top, there will be a short introduction that will display the characteristics the project will provide.

Quick Templates Access

There should be good templates for an ID card that would be displayed.

Tips

The tips related to the generation of ID cards will be displayed.

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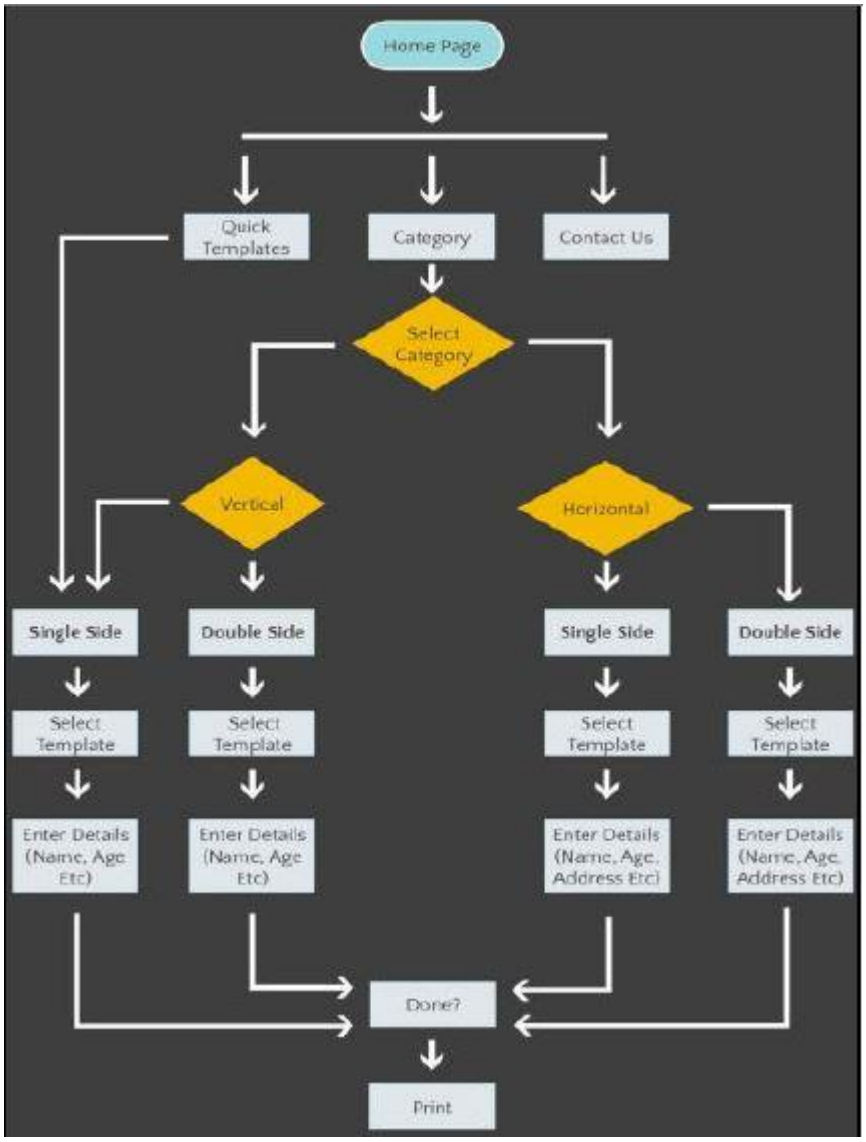

Figure 1 Project Description

1.5 Summary

In the chapter, we have discussed the introduction of ID cards, the traditional process for making ID cards in government degree college Wari, and a proposed solution for the concern issue.

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CHAPTER – 2:

REQUIREMENT ENGINEERING

It relates to the engineering design process's steps for identifying, capturing, and managing requirements. Requirement engineering is the correct way to know about the need of the user, feasibility study, requirements analysis, requirements validation, proposing a good solution, and managing all the requirements to convert them into a functional system. So it is clear that it is the step-by-step application of techniques, tools, and methodologies to estimate the behavior of the proposed project. The phenomenon of elicitation of stakeholder needs and then turning them into a proper set of requirements, which will be the foundation for ensuring the developmental process, is termed requirements engineering. Two goals of the requirements engineering process include problem-making and an accurate, reliable, reasonable, and efficient solution. This chapter lists the typical requirements engineering approaches currently in use, mainly concentrating on the fundamentals. Requirements engineering techniques are methods for transforming real-world issues into solutions for the digital world. Each mode uses a distinct process to develop the system specification as the solution, each with its specialized way of thinking about the real-world problem.

2.1Use case scenarios

Use case scenarios are often used in software development as part of the functional requirement documents/plan, but they aren't only for developers. They illustrate how users could interact with a system to achieve a goal. In requirement analysis, we examine the needs to comprehend them as the foundation for obtaining the desired functionality while maintaining the system's quality, dependability, and efficiency. We discover the user's requirements through requirement analysis. This process involves gathering the needs from

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various sources. The analyst does the requirement analysis. It is a very significant and vital phase in the software development process since it is the foundation of software development. The needs are gathered in this stage from several sources, including.

2.1.1 Use case scenarios: user Registration and login

A person who has already registered with a website, application, or other system is known as a registered user. In order to prove their identity, registered users typically provide the system with credentials (such as a username, email address, and password). The use case below illustrates a user registering for the first time in the application, the response provided, and the exception where the details are incorrect. This use case also shows how to log in after signing up, except when the information is wrong.

Name Use case	Registering
Use case identifier	UC#2
Actor(s)	Scholars
Account	Students register for a course
Precondition	They have an account and security key to log in
Postcondition	Student’s registration is achievement full.
Basic course of Actions	
1. Students desire to sign up for the course 2. Students Register on the page. 3. Complete the page's needed fields. 4. The student's prior grade was higher than or equal to the passing grade. 5. Successful Registration 6. Students get registration messages in 6. 7. Use case ends.	

2.1.2Use case scenarios: using Dashboard

A dashboard's primary purpose is to display a thorough overview of data from many sources. Dashboards help track, measure, and evaluate pertinent data in essential areas.

Login, Registration, and Dashboard steps

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1. Open the home page.
2. A login module prompt will appear for the user.
3. In order to access more features of the system, users must have login information.
4. The user will be sent to the Dashboard after entering their valid credentials.
5. They can generate both new and current cards.
6. They may also edit and change cards as well as delete them.

2.2Functional Requirement

Software development and system engineering define the desired functioning of programmers and systems or function requirements. Software-driven electronics or a combination of software and electronic hardware might make up the system under system engineering.

- Users can search the entire original collection of databases or only a portion of them.
- Users can access the system to view the information and give input.
- The developer can only do modifications to the system.
- Every comment is gathered from the user's email address. Each user must supply a password as well as a unique identity.

2.3Non -Functional Requirement

The system attributes defined by non-functional requirements (NFRs), such as security, reliability, performance, maintainability, scalability, and usability, serve as limitations or restrictions on the system's design throughout the many backlogs.

Usability

The system provides a help and support menu in each interface so that the user may interact with the system.

Safety

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Given that our website is web-based and operates online. Threats such as viruses and malware will always exist, but they may be reduced by using a new browser, installing anti-spyware software, and staying up to date with Microsoft Windows Defender's most recent definition.

Performance

Only people with some background on the internet, web browsing, and knowledge of other social networks will utilize our website. They don't need to worry about using our website.

Availability

The system should always be available for access 24 hours. Seven days a week.

Efficiency

We developed an efficient website because it can serve many users simultaneously and will respond to every user very efficiently.

Reliability

Our website is dependable since the programmer consistently exhibits the same behavior and does not alter its result at every step. This chapter has covered every aspect of the prerequisites. In order to obtain the necessary needs, we collected the requirements and then examined those requirements. The completed specifications were then verified following that.

2.4Data base Requirement

When referencing data requirement data in documents, it should be done in business language and, if possible, by using approved standard business terms. The data requirement

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definition establishes the process used to identify priorities and precisely formulate and validate the data needed to achieve business objectives.

2.5 Summary

This chapter includes system requirements and the Engineering of the system. It provides user registration for the project and designs: using Dashboard, Functional Requirements, and Database Requirements.

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CHAPTER – 3:

DESIGNING

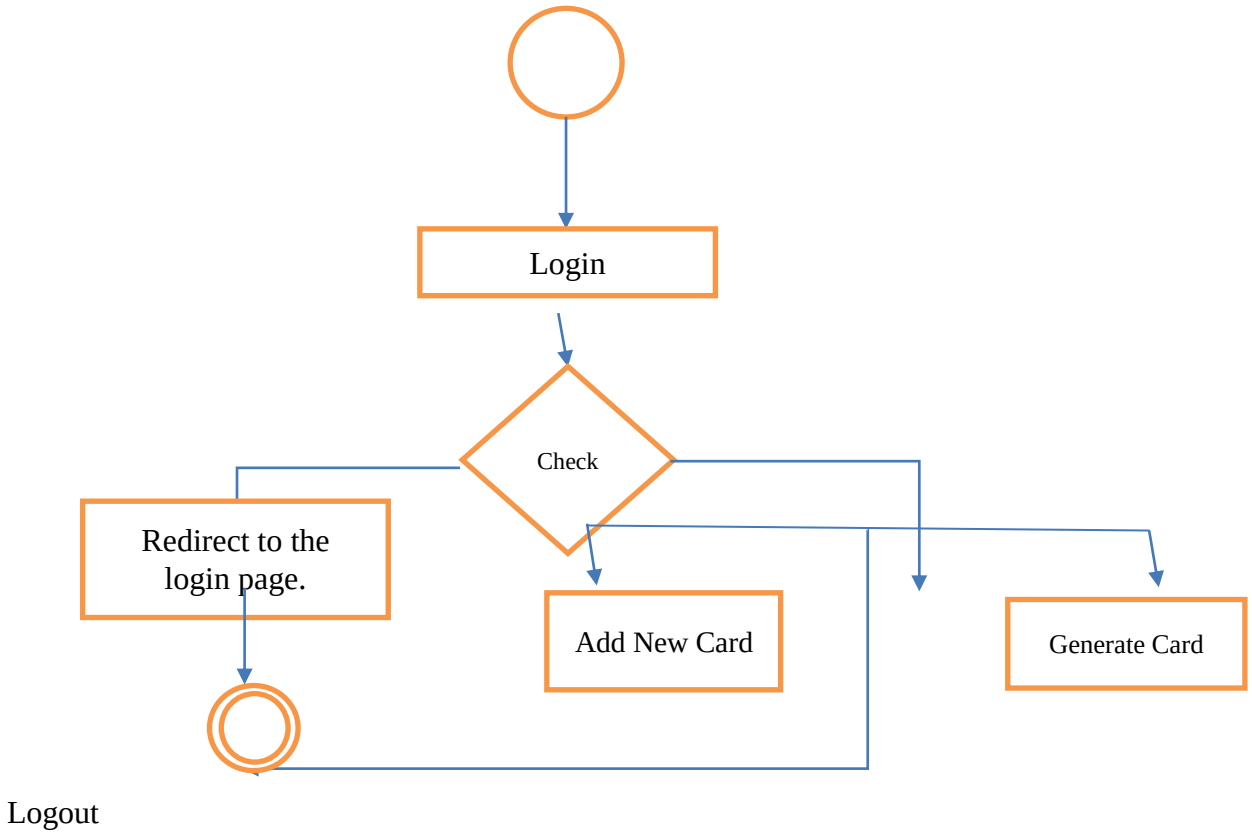
The procedure in which we imagine and define a software solution to one or more than one problems is termed software design. Software requirement analysis (SRA), a step in the software development process that includes a specification list for software engineering, is one of the critical elements of software design.

3.1Data Flow diagram (DFD)

A data flow diagram is a graphical or visual depiction that uses a defined set of systems and notation to explain how data is moved through a business transaction. They are frequently part of formal methodologies like structured system analysis and method (SSADM).

3.2Activity diagram

Activity Diagram



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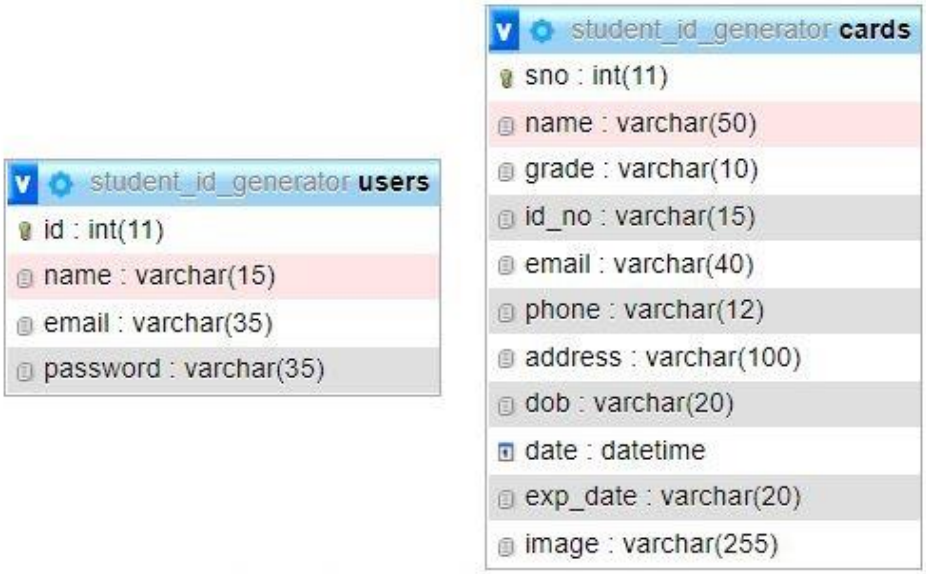
3.3Entity Relationship Diagram (ERD)

An Entity Relationship Model (ERM), often referred to as an Entity Relationship Diagram (ERD), is a visual representation that shows the relationships between entities such as places, an individual, ideas, or events inside an (IT) system.



ERD: **Student ID Card Generator System**

3.4Summary

This chapter includes system Designing and DFD of the system. It uses an Activity diagram for the project and an Entity Relationship Diagram.

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CHAPTER – 4:

IMPLEMENTATION

Any document that may be used to confirm features of a person's personality qualifies as an identity document. It is typically referred to as an identification card if issued in the shape of a compact, typically standard-sized Card (IC). Informal papers may be needed in nations without official identification documents.

Driving licenses can be used in many countries as a means of proving identity in the lack of a formal identity document, while other nations need to recognize driving licenses as identification frequently because they do not expire there and can be readily falsified. Passports are recognized as valid forms of identification in most nations.

Foreign nationals are often required to have a passport or, rarely, a national identity card.

4.1 Tools used for implementation

The tools used for the development of the system are enlisted below:

- Word 2016 in MS (Report writing)
- My SQL database server
- Languages used in programming: PHP, CSS, HTML,
- Microsoft Visio Photoshop (Pictures) (forming charts)

4.1.1Integrated development environment (IDE)

Eclipse for programming in Java Android Studio and Microsoft Visual Studio for creating mobile applications. Some popular IDEs include PyCharm for Python programming and RStudio for R scripts. An integrated development environment (IDE) is a piece of software that aids in programmers' productive creation of software code.

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4.1.2Personal Home Page (PHP)

Hypertext Preprocessor (PHP) is an accepted open-source programming language. The server is where PHP programmers are run. PHP, in most cases, is used for developing websites; sometimes, server-side scripting is used for the general-purpose programming language. The PHP Development RasmuLerdorf developed it in 1994; nowadays, different teams produce the reference implementation.

4.1.3 Hypertext Markup Language (HTML)

(HTML) stands for Hyper Text Markup Language. The standard for web pages is HTML. HTML explains how a web page is organized. HTML has many elements that instruct browsers how to display material.

The industry-standard markup language for developing online pages and web applications is HTML. JavaScript and Cascading Style Sheet (CSS) make up the trio of foundational technologies underlying the World Wide Web. Internet browsers get HTML initially featured cues for the document's design and semantically explain the structure of a web page.

4.1.4 Cascading Style Sheets (CSS)

Cascading Style Sheets (CSS) is a style sheet language used to specify how a document written in HTML or XML, including XML dialects like SVG MathML or XHTML, should be presented. CSS specifies how elements should be represented on screen, paper, speech, or other mediums.

4.1.5 JavaScript

HTML and CSS are the languages that offer structure and style to web pages, whereas JavaScript is a text-based computer language used both on the client and server sides. JavaScript added interactive elements to websites that keep users interested.

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JavaScript is a dynamic programming language used for creating websites and many other things. You can add active website features that you couldn't achieve with only HTML and CSS.

4.2 Working of the project: Screen Slide of the project

Home / Index Page

The Main page looks like this

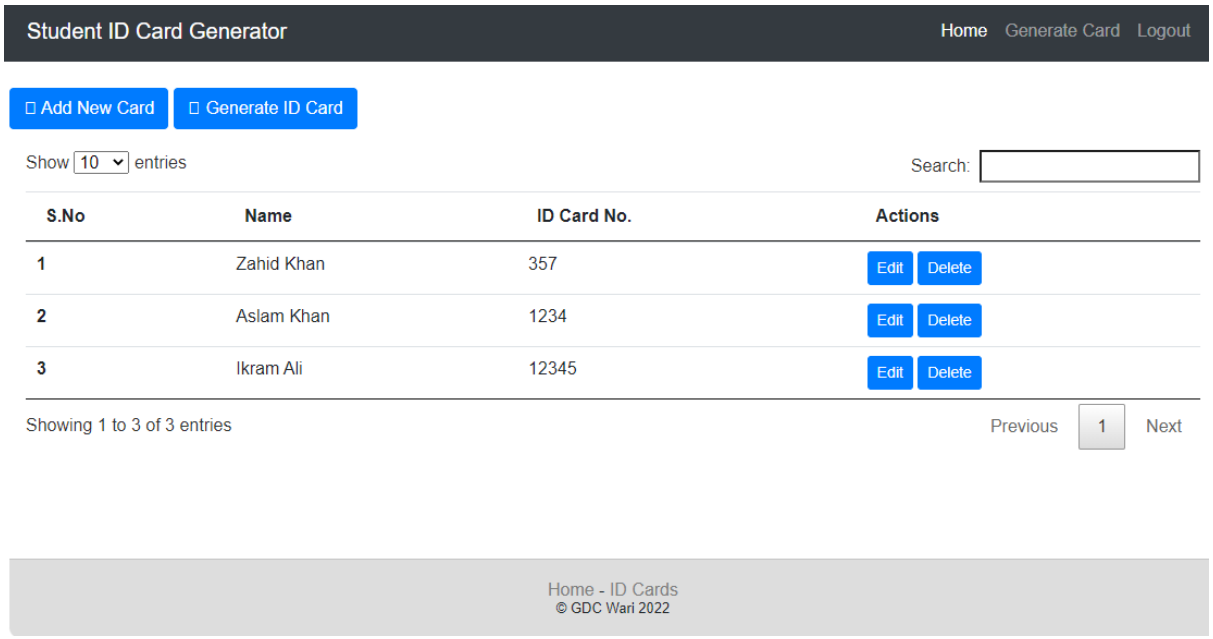

Figure 2Index Page

Login Page

When you click on the login button following page opens

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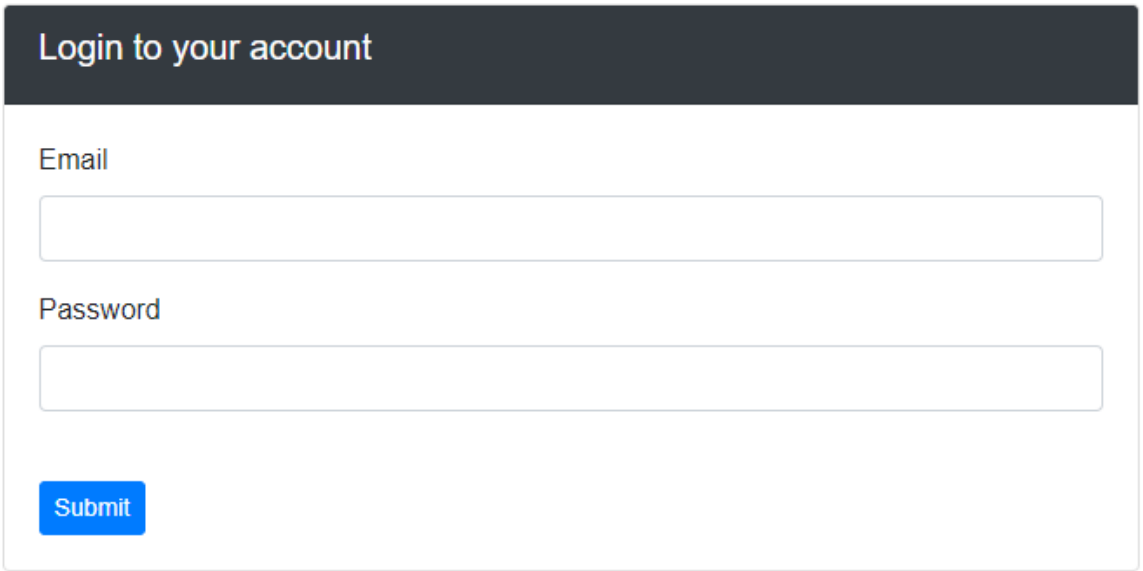

Figure 3Login Page

Login Page

When you click on the login button following page opens

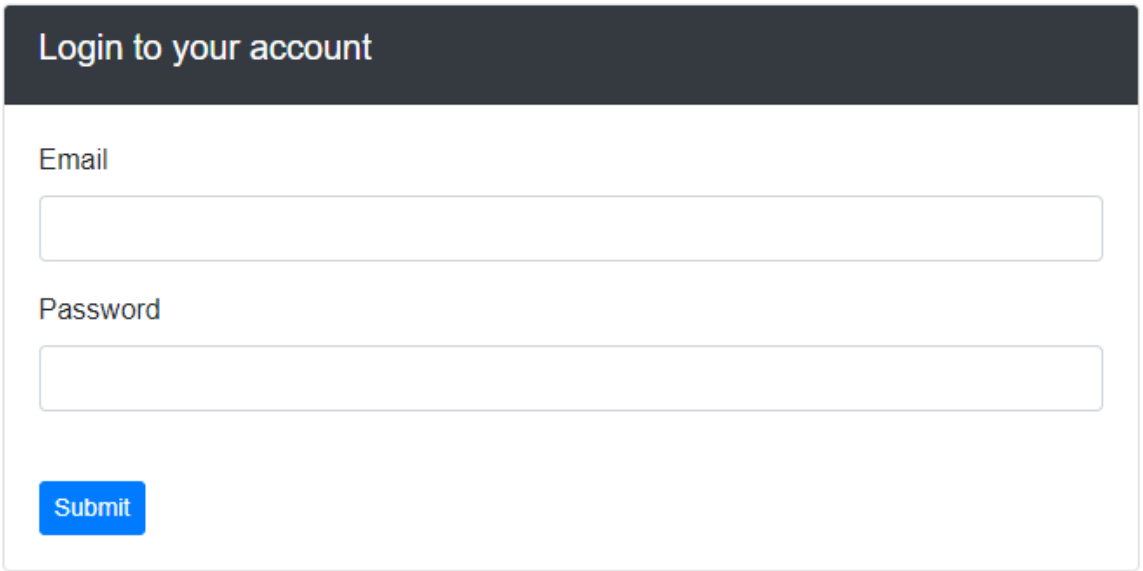

Figure 4Login Page

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Adding new Card

When you click the Add New Card button, the following interface opens to create a new card.

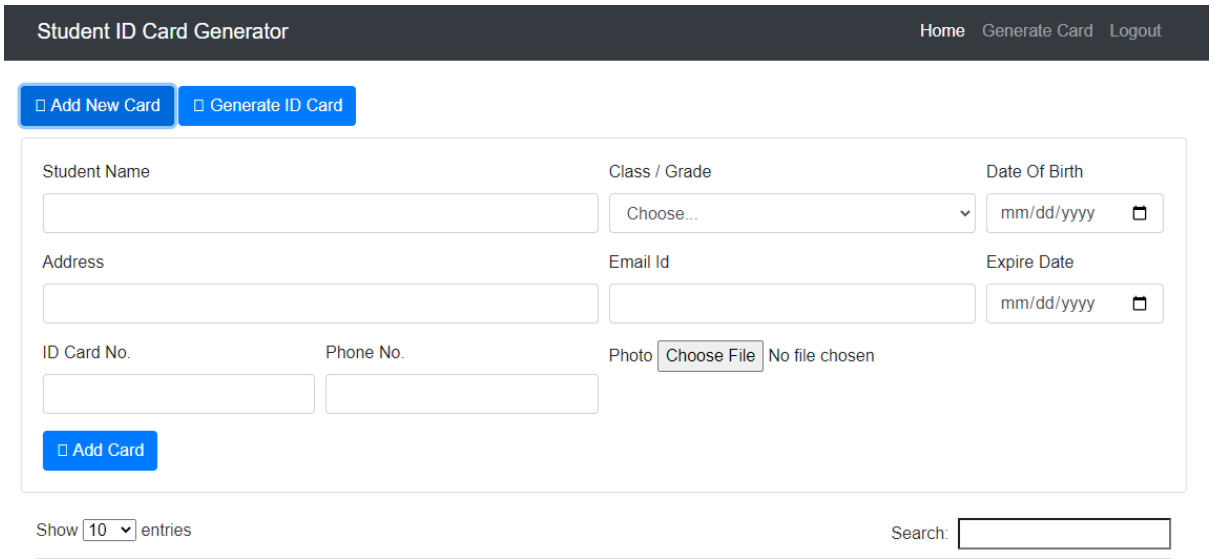

Figure 5Registration Page5.4 Generating a new Card

By entering the Card ID in the input field. Their Card will be generated, and you can download and print it.

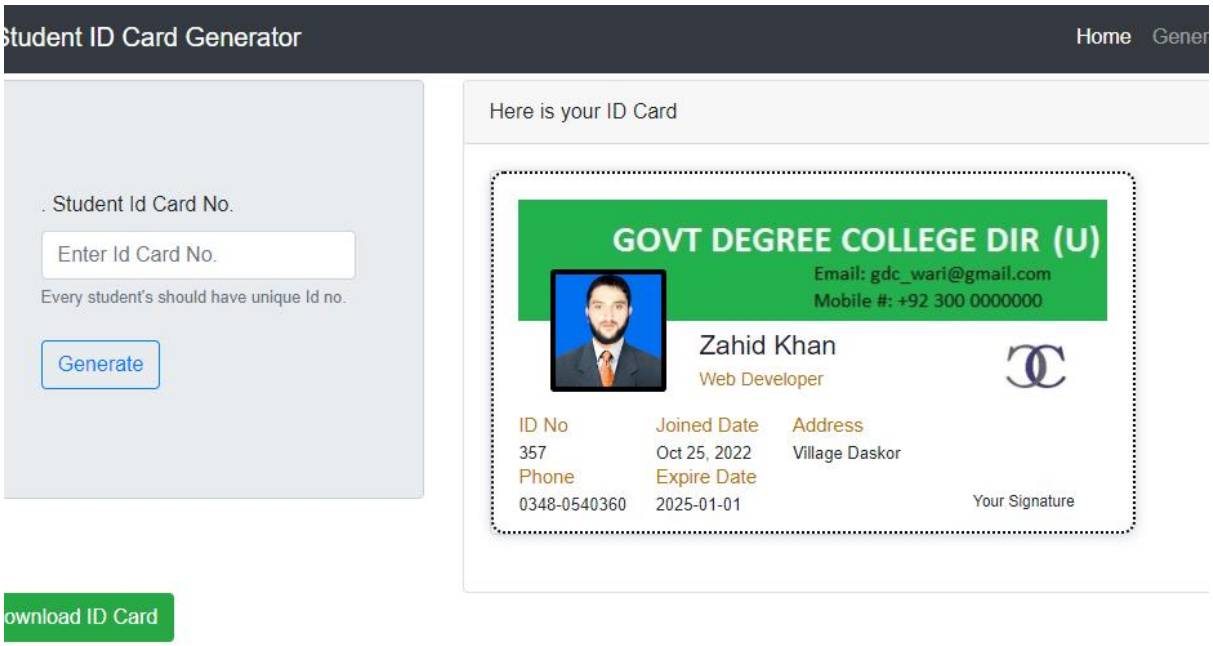

Figure 6 Generating new ID Card Figure 7Update Profile Details

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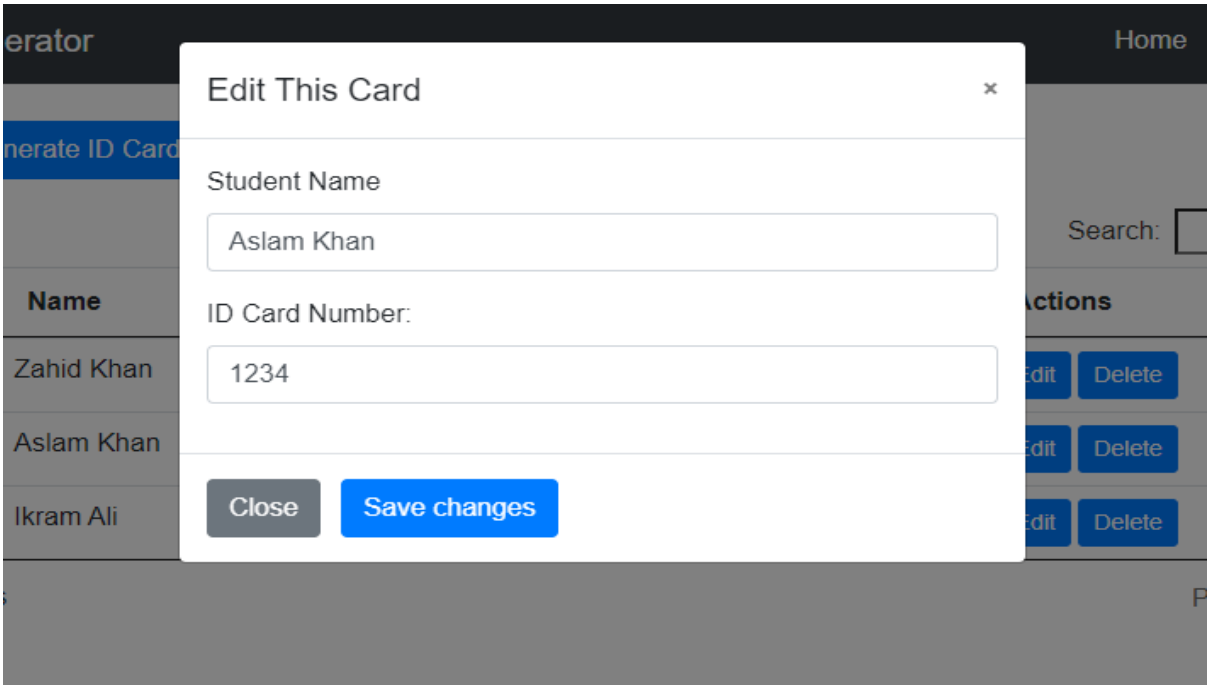
Update Card Details

Users can change their card details on most



4.3 Summary

The chapter includes system Implementation and Tools of the system. It provides user IDE, PHP, HTML, and CSS for the project and JavaScript and Summary.

Approval.

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CHAPTER – 5:

CONCLUSION

5.1 Conclusion and future work

The ID card generator system will be user-friendly and a practical application for government degree college Wari students. The traditional card generator system used for students' ID cards makes it very expensive and time-consuming. They already used the design and also needed help in providing details. The proposed project will reduce the cost also in printing the cards. It provides an error-free system that will eliminate all the mistakes in traditional cards generating systems. The proposed project ID card generator system for the government degree college Wari will automatically create identity cards for the students of the government degree college Wari district Dir Upper, which will show efficiency in terms of time and cost. As government degree college wari during the fall semester's intake of large numbers of students bound to do a mistake. This mistake is elementary and will be significant, such as writing down the wrong details. All these errors will be removed using the proposed project id generator system for the government degree college Wari. The proposed project should be enhanced in the future in terms of an employee. Administration staff card generating feature.

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